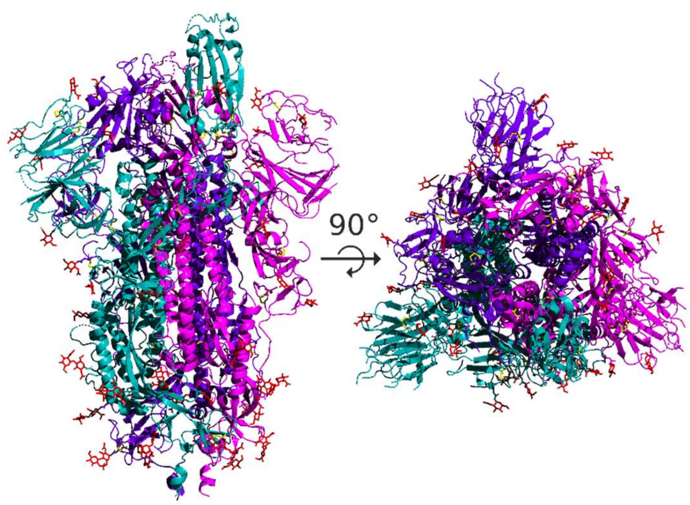
SARS-CoV-2 Spike Protein Structure Analysis

This challenge is adapted from a longer module that is part of my *Biological Modeling* project. If you enjoy this material, check out that course and its accompanying textbook at https://biologicalmodeling.org.

In previous assignments, we assembled and annotated the SARS-CoV-2 genome, and we then examined how this genome is changing as the virus mutates within the human population. We saw that some regions of the genome are more variable than others compared to the SARS-CoV genome; we will focus on one of these regions, the gene encoding the spike protein.

In this assignment, we will place ourselves in the shoes of early SARS-CoV-2 researchers studying the new virus in January 2020 after the virus's genome was published. We now ask ourselves two questions. First, can we use the virus's genome to determine the structure of its spike protein? Second, once we know the structure of the SARS-CoV-2 spike protein, how does its structure and function differ from the same protein in SARS-CoV?

The SARS-CoV and SARS-CoV-2 spike protein is a **homotrimer**, meaning that it is formed of three identical chains, each one translated from the corresponding region of the coronavirus's genome.



The SARS-CoV-2 spike protein has a **receptor binding domain (RBD)** located on the S1 subunit of the spike protein that is responsible for interacting with the human ACE2 enzyme; the rest of the protein does not come into contact with ACE2.

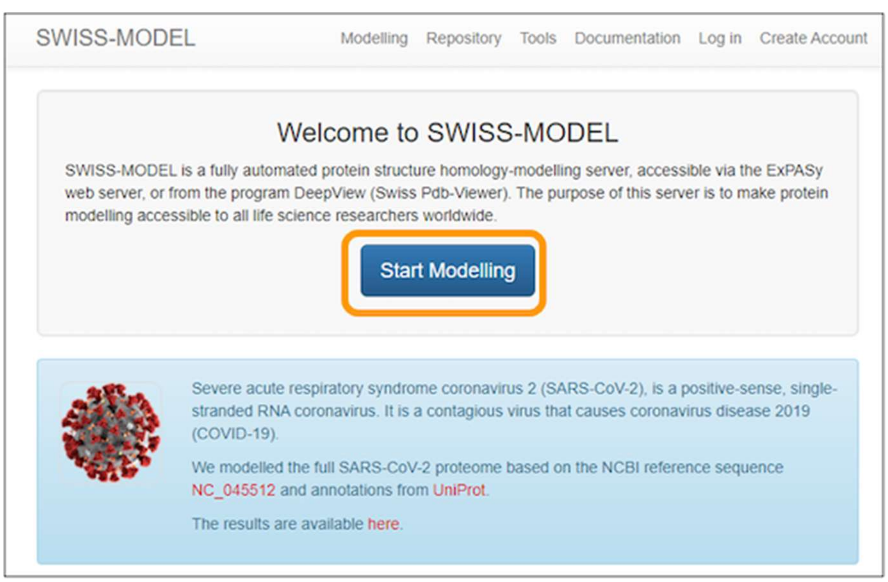
In this assignment, we will predict the structure of the SARS-CoV-2 spike protein from its amino acid sequence. We will then learn how to compare our predicted structure against the known experimental structure, and use this approach to compare the SARS-CoV-2 spike protein's structure against that of SARS-CoV, the virus that caused the original 2003 outbreak.

Note: some parts of this challenge require running software that can take several hours to finish running, so make sure to give yourself plenty of time since you won't be able to complete this challenge in one sitting.

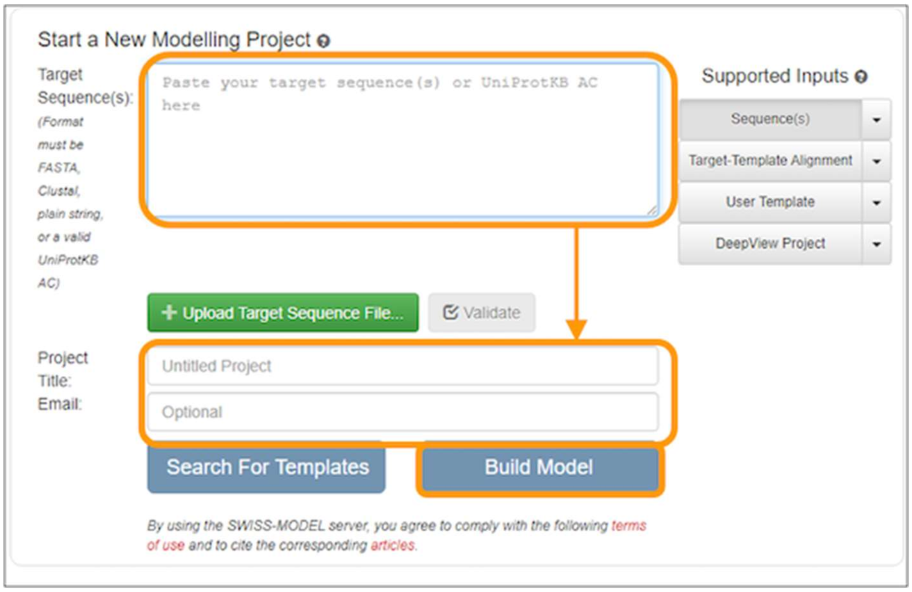
In ***a priori*** **structure prediction**, we attempt to reconstruct the (tertiary) structure of a protein directly from its amino acid sequence, without taking any known structures into account. Unfortunately, the algorithms implementing *a priori* structure prediction take a very long time to run; in fact, some publicly available *a priori* structure prediction software caps the length of the protein.

Since the SARS-CoV-2 spike protein gene encodes a polypeptide chain with 1281 amino acids, we will instead turn our attention to **homology modeling**, in which we use a database of experimentally validated structures to increase the accuracy and efficiency of structure prediction.

There are several publicly available resources for homology modeling; we will use a web server called SWISS-MODEL. To run SWISS-MODEL, first download the sequence of the spike protein chain: [SARS-CoV-2 spike protein chain](). Next, go to the main [SWISS-MODEL website]() and click *Start Modelling*.



On the next page, copy and paste the sequence into the *Target Sequence(s):* box. Name your project and enter an email address to get a notification of when your results are ready. Finally, click *Build Model* to submit the job request. Note that you do not need to specify that you want to use the SARS-CoV spike protein as a template because the software will automatically search for a template for you (and find this template).



Important note: your results may take between an hour and a day to finish depending on how busy the server is. When you receive an email notification, follow the link provided and you can download the final models produced.

When SWISS-MODEL finishes, visit the results page. You will see a collection of structure models produced as a result of the modeling process. By default, an image of the "model 1" predicted protein is shown on the right of the page. Feel free to explore this model by clicking to rotate it.

Exercise: SWISS-MODEL returned multiple results. Give and explain two reasons why returning multiple results would be helpful for researchers.

> One reason we may want to have multiple similar results is that more observations means higher confidence that the structure is true and accurate. The more often we see the same helix, loop, or folding, the more likely it is that the sequence actually takes that shape.
>
> Sequences can also just have many structural differences with similar low energy. While the secondary structure is usually similar, differences can arise in the tertiary

structure. Proteins have conformations with different functions, so having information on each of them can help researchers investigate structures before and after substrate binding or DNA binding.

SWISS-MODEL provides metrics to attempt to determine the quality of each of the models that it predicts. We will allow you to explore these metrics at the SWISS-MODEL documentation but will mention that one of the metrics is **QMEAN**, which attempts to quantify how well a modeled structure compares against what we would “expect” to see in experimental structures. SWISS-MODEL furthermore colors the predicted structure on a spectrum based on QMEAN quality scores, with blue regions representing high similarity, and red regions representing lower similarity. By clicking on the image of different models on the left, you can see how the models differ.

In the meantime, we want to save the models that we produced so that they can be used in other analyses. To do so, click the download icon near the top of the page and save the models to your computer.

Exercise: For each of the models produced by SWISS-MODEL, explain what template was used, which organism it came from, and what its sequence identity is with the SARS-CoV-2 molecule.

Model 1 uses a cryo-EM structure of Bat SARS-like coronavirus WIV1 spike glycoprotein (8wly.1.A) and has sequence identity 78.82%.

Model 2 uses a cryo-EM structure of Bat SARS-like coronavirus RsSHC014 spike glycoprotein (8wlu.1.A) and has sequence identify 79.07%.

Model 3 uses a cryo-EM structure of SARS-CoV-2 Omicron JN.1 spike protein (8y5j.1.A) and has sequence identity 94.87%.

Model 4 uses a structure of apo SARS-CoV-2 spike protein with one RBD up (8h3d.1.A) and has sequence identity 99.60%.

Model 5 uses a cryo-EM structure of the WIV1 S-hACE2 complex from Bat SARS-like coronavirus WIV1 (8wlz.1.C) and has sequence identity 78.82%.

Model 6 uses a severe acute respiratory syndrome coronavirus 2 BA.2.86 spike protein in complex with ACE2 (8yzc.1.A) and has sequence identify 94.08%.

Model 7 uses an EM structure of BANAL-20-236 S1 in complex with R. Affinis ACE2 from Rhinolophus affinis/unclassified Coronavirinae (8hxk.1.B) and has sequence identity 90.61%.

Comparing a predicted spike protein structure against SARS-CoV-2

Our prediction of the SARS-CoV-2 spike protein mimics the work of many researchers in January 2020, which included the contributions of volunteers' computers from around the world.

Meanwhile, other scientists were working on verifying the structure of the protein experimentally. On February 25, 2020, researchers from the Seattle Structural Genomics Center for Infectious Disease deposited the result of a cryo-EM experiment determining the structure of the spike protein to the Protein Data Bank (PDB) as entry 6vxx. Cryo-EM is described in the following YouTube video if you are interested: https://www.youtube.com/watch?v=Qq8DO-4BnIY

Our goal is to compare the models that SWISS-MODEL predicted against the validated structure of SARS-CoV-2. To do so, we will need to have a file format that represents the tertiary structure of a protein. We will use **.pdb format**, which is illustrated in the figure below. Each atom in the protein is labeled according to several different pieces of information, including:

1. the element from which the atom derives;
2. the amino acid in which the atom is contained;
3. the chain on which this amino acid is found;
4. the position of the amino acid within this chain; and
5. the 3D coordinates (*x*, *y*, *z*) of the atom in angstroms (10-10 meters).

							Coordinates		
		Index	Element	Amino acid	Chain	Position	*x*	*y*	*z*
2159	ATOM	1	N	ALA	A	27	171.646	251.874	224.877
2160	**ATOM**	**2**	**CA**	**ALA**	**A**	**27**	**172.298**	**252.181**	**223.613**
2161	ATOM	3	C	ALA	A	27	173.530	251.298	223.427
2162	ATOM	4	O	ALA	A	27	174.195	250.943	224.405
2163	ATOM	5	CB	ALA	A	27	172.700	253.664	223.554
2164	ATOM	6	N	TYR	A	28	173.816	250.939	222.166
2165	**ATOM**	**7**	**CA**	**TYR**	**A**	**28**	**174.968**	**250.129**	**221.763**
2166	ATOM	8	C	TYR	A	28	175.652	250.729	220.561
2167	ATOM	9	O	TYR	A	28	175.009	251.379	219.736
2168	ATOM	10	CB	TYR	A	28	174.546	248.703	221.426
2169	ATOM	11	CG	TYR	A	28	174.049	247.932	222.586
2170	ATOM	12	CD1	TYR	A	28	172.752	248.072	223.009
2171	ATOM	13	CD2	TYR	A	28	174.897	247.067	223.225
2172	ATOM	14	CE1	TYR	A	28	172.304	247.348	224.080
2173	ATOM	15	CE2	TYR	A	28	174.455	246.338	224.291
2174	ATOM	16	CZ	TYR	A	28	173.161	246.477	224.723
2175	ATOM	17	OH	TYR	A	28	172.710	245.746	225.795

Open the folder that you downloaded from SWISS-MODEL. You will see a directory corresponding to each of the models that the software predicted. Within each directory, there is a *model.pdb* file that can be opened with a text editor.

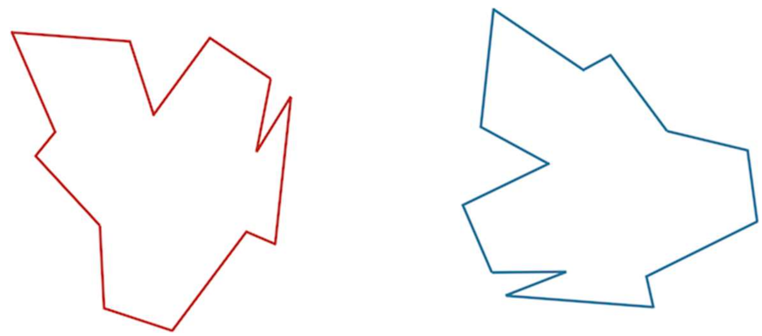
Exercise: Provide the first ten lines corresponding to atomic positions of your model 1.

```
ATOM    1  N   GLN A  33     95.482 112.255 190.203  1.00  0.36          N
ATOM    2  CA  GLN A  33     96.859 111.878 189.740  1.00  0.36          C
ATOM    3  C   GLN A  33     97.680 113.121 189.515  1.00  0.36          C
ATOM    4  O   GLN A  33     97.095 114.198 189.514  1.00  0.36          O
ATOM    5  CB  GLN A  33     96.727 111.068 188.429  1.00  0.36          C
ATOM    6  CG  GLN A  33     96.060 109.680 188.604  1.00  0.36          C
ATOM    7  CD  GLN A  33     96.960 108.779 189.453  1.00  0.36          C
ATOM    8  OE1 GLN A  33     98.144 108.767 189.135  1.00  0.36          O
ATOM    9  NE2 GLN A  33     96.424 108.104 190.491  1.00  0.36          N
ATOM   10  N   CYS A  34     99.023 113.009 189.367  1.00  0.49          N
```

Comparing two shapes with the Kabsch algorithm

We now would like to compare the known SARS-CoV-2 protein structure against a structure predicted by a model, which is an example of the more general problem of comparing two shapes.

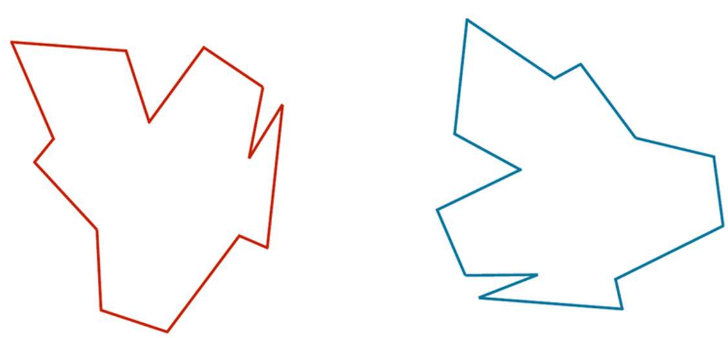
Consider the two shapes in the figure below. How similar are they?



If you think you have a good handle on comparing the above two shapes, then it is because humans have very highly evolved eyes and brains. But training a computer to detect and differentiate objects is more difficult than you think!

We would like to develop a distance function $d(S, T)$ quantifying how different two shapes S and T are. If S and T are the same, then $d(S, T)$ should be equal to zero; the more different S and T become, the larger d should become.

You may have noticed that the two shapes in the preceding figure are, in fact, identical. To demonstrate that this is true, we can first move the red shape to superimpose it over the blue shape, then flip the red shape, and finally rotate it so that its boundary coincides with the blue shape, as shown in the animation below. In general, if a shape S can be translated, flipped, and/or rotated to produce shape T, then S and T are the same shape, and so $d(S, T)$ should be equal to zero. The question is what $d(S, T)$ should be if S and T are not the same shape.



Our idea for defining $d(S, T)$, then, is first to translate, flip, and rotate S so that it resembles T "as much as possible" to give us a fair comparison. Once we have done so, we will devise a metric to quantify the difference between the two shapes that will represent $d(S, T)$.

We first translate S to have the same **center of mass** (or **center of mass**) as T. The center of mass of S is found at the point (xS, yS) such that xS and yS are the respective averages of the x-coordinates and y-coordinates on the boundary of S.

The center of mass of some shapes, like the arc in the preceding example, can be determined mathematically. But for irregular shapes, we will first sample n points from the boundary of S and then estimate xS and yS as the average of all the respective x- and y-coordinates from the sampled points.

After finding the center of mass of the two shapes S and T that we wish to compare, we translate S so that it has the same center of mass as T. We then wish to find the rotation of S, possibly along with a flip as well, that makes the shape resemble T as much as possible.

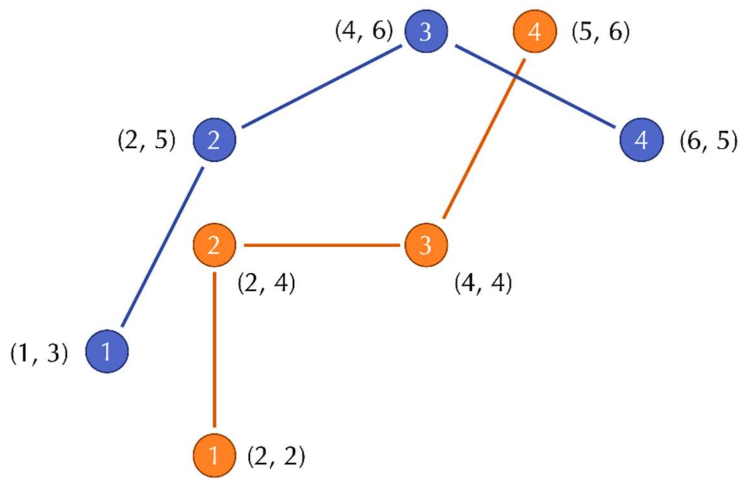
Imagine that we have found the desired rotation; we are now ready to define $d(S, T)$ in the following way. We sample n points along the boundary of each shape, converting S and T into **vectors** $\mathbf{s} = (s1, \ldots, sn)$ and $\mathbf{t} = (t1, \ldots, tn)$, where si is the i-th point on the boundary of S. The **root mean square deviation (RMSD)** between the two shapes is the square root of the average squared distance between corresponding points in the vectors,

$$\mathrm{RMSD}(\mathbf{s}, \mathbf{t}) = \sqrt{\frac{1}{n} \cdot \left(d(s_1, t_1)^2 + d(s_2, t_2)^2 + \cdots + d(s_n, t_n)^2\right)}$$

In this formula, $d(si, ti)$ is the distance between the points si and ti.

For an example two-dimensional RMSD calculation, consider the figure below, which shows two shapes with four points sampled from each. (Note: for simplicity, the shapes do not have the same center of mass.)In this formula, $d(si, ti)$ is the distance between the points si and ti.

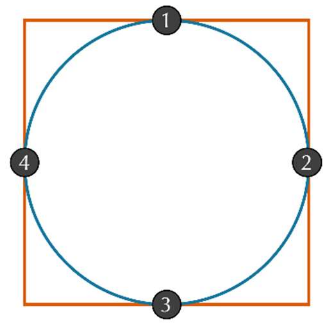
For an example two-dimensional RMSD calculation, consider the figure below, which shows two shapes with four points sampled from each. (Note: for simplicity, the shapes do not have the same center of mass.)



The distances between corresponding points in this figure are equal to $\sqrt{2}$, 1, 4, and $\sqrt{2}$. As a result, we compute the RMSD as

$$\begin{aligned}\text{RMSD}(\mathbf{s}, \mathbf{t}) &= \sqrt{\frac{1}{4} \cdot (\sqrt{2}^2 + 1^2 + 2^2 + \sqrt{2}^2)} \\ &= \sqrt{\frac{1}{4} \cdot 9} \\ &= \sqrt{\frac{9}{4}} \\ &= \frac{3}{2}\end{aligned}$$

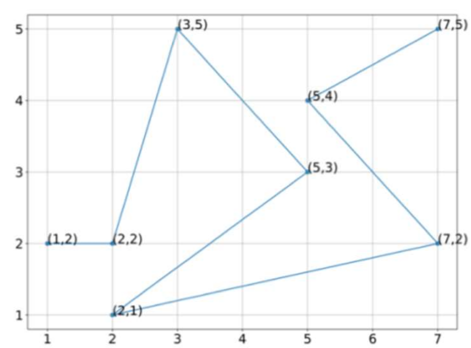
Even if we assume that two shapes have already been overlapped and rotated appropriately, we still should ensure that we sample enough points to give a good approximation of how different the shapes are. Consider a circle inscribed within a square, as shown in the figure below. If we happened to sample only the four points indicated, then we would sample the same points for each shape and conclude that the RMSD between these two shapes is equal to zero. Fortunately, this issue is easily resolved by making sure to sample enough points from the shape boundaries.

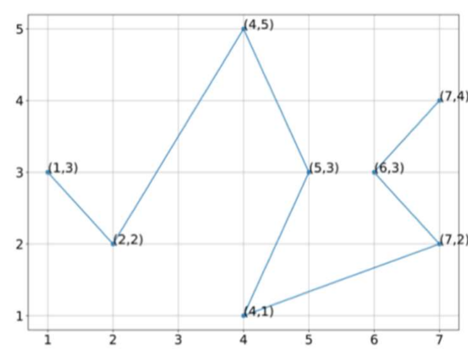


However, we have still assumed that we already rotated (and possibly flipped) *S* to be as "similar" to *T* as possible. In practice, after superimposing *S* and *T* to have the same center of mass, we would like to find the flip and/or rotation of *S* that *minimizes* the RMSD between our vectorizations of *S* and *T* over all possible ways of flipping and rotating *S*. It is this minimum RMSD that we define as *d*(*S*, *T*).

The best way of rotating and flipping *S* so as to minimize the RMSD between the resulting vectors *s* and *t* can be found with a method called the **Kabsch algorithm**. Explaining this algorithm requires some advanced linear algebra and is beyond the scope of our work but is described here.

Exercise: Using the vectorization of the figures indicated below, estimate the center of mass of each of these two protein structures. Show your work.





xS: (1+2+2+3+5+5+7+7) / 8 = 4

yS: (1+2+2+2+3+4+5+5) / 8 = 3

Center of mass(S) = (xS, yS) = (4, 3)

xT: (1+2+4+4+5+6+7+7) / 8 = 4.5

yT: (1+2+2+3+3+3+4+5) / 8 = 2.875

Center of mass(T) = (xT, yT) = (4.5, 2.875)

Exercise: Without rotating these structures, align them so that they have the same center of mass, and determine the RMSD between the two structures for the vectorizations shown. Show your work.

Translation: T-S = (xS - xT, yS - yT) = (4.5 - 4, 2.875 - 3) = (0.5, -0.125) = (xS*,yS*)

xS1* = xS1 + xS* = 1 + 0.5 = 1.5	yS1* = yS1 + yS* = 2 - 0.125 = 1.875
xS2* = 2 + 0.5 = 1.5	yS2* = 2 - 0.125 = 1.875
xS3* = 2 + 0.5 = 2.5	yS3* = 1 - 0.125 = 0.875
xS4* = 3 + 0.5 = 3.5	yS4* = 5 - 0.125 = 4.875
xS5* = 5 + 0.5 = 5.5	yS5* = 3 - 0.125 = 2.875
xS6* = 5 + 0.5 = 5.5	yS6* = 4 - 0.125 = 3.875
xS7* = 7 + 0.5 = 7.5	yS7* = 2 - 0.125 = 1.875
XS8* = 7 + 0.5 = 7.5	yS8* = 5 - 0.125 = 4.875

Pair	x	y	distance^2
s1*-t1	1.5 - 1 = 0.5	1.875 - 3 = -1.125	0.5^2 + (−1.125)^2 = 1.516
s2*-t2	1.5 – 1 = 0.5	1.875 - 2 = -0.125	0.5^2 + (−0.125)^2 = 0.266
s3* - t3	-1.5	-0.125	(-1.5)^2 + (-0.125)^2 = 2.266
s4* - t4	-0.5	-0.125	(-0.5)^2 + (-0.125)^2 = 0.266
s5* - t5	0.5	-0.125	(0.5)^2 + (-0.125)^2 = 0.266
s6* - t6	-0.5	0.875	(-0.5)^2 + (0.875)^2 = 1.016
s7* - t7	0.5	-0.125	(0.5)^2 + (-0.125)^2 = 0.266
s8* - t8	0.5	0.875	(0.5)^2 + (0.875)^2 = 1.016

RMSD = sqrt(sum(distance^2) / n)

= sqrt((1.516 + 0.266 + 2.266 + 0.266 + 0.266 + 1.016 + 0.266 + 1.016) / 8)

= **0.927**.

Comparing our structure against an experimentally verified structure

We will now use the Kabsch algorithm to compare our models against the verified structure of SARS-CoV-2. Our featured software resource is **ProDy**, an open-source Python package that allows users to perform protein structural dynamics analysis. Its flexibility allows users to select specific parts or atoms of the structure for conducting normal mode analysis and structure comparison.

To get started, make sure that you have the following software resources installed in order to use Jupyter Notebook. You can also access Jupyter Notebook by using Anaconda Individual Edition (we suggest downloading the graphical installer).

- Python (3.5 or later)
- ProDy
- NumPy
- Biopython
- IPython
- Jupyter
- Matplotlib

Next, create a directory somewhere on your computer, and move the five .pdb files that you obtained from running SWISS-MODEL to this directory. Name these files *swiss_model_1.pdb*, *swiss_model_2.pdb*, etc.

You can find a completed version of our notebook here. You may also like to follow along below as we go through this notebook.

Start Jupyter notebook by either opening Anaconda or typing jupyter notebook into a command line terminal.

Before you run the Jupyter notebook file provided, we will explain each of the components of this file. First, we have

*from prody import **

which imports the Prody package and can use all functions from this package.

The first built-in ProDy function that we will use, called *parsePDB*, parses a protein structure in *.pdb* format. Let's parse in one of our models we obtained from homology modeling of the SARS-CoV-2 Spike protein, *swiss_model_1.pdb*.

In[#]: struct1 = parsePDB('swiss_model_1.pdb')

To compare our model against the determined protein structure of the SARS-CoV-2 Spike protein, we will parse the protein with PDB ID *6vxx*.

In[#]: struct2 = parsePDB('6vxx')

Because the *.pdb* extension is missing, this command will prompt the console to search for *6vxx*, the SARS-CoV-2 Spike protein, from the PDB and download the *.pdb* file into the current directory. It will then save the structure as the variable *struct2*.

With the protein structures parsed, we can now match chains. To do so, we use a built-in function *matchChains* with a sequence identity threshold of 75% and an overlap threshold of 80% is specified (the default is 90% for both parameters). The following function call stores the result in a 2D array called *matches*. In this array, *matches[i]* denotes the *i*-th match found between two chains that are stored as *matches[i][0]* and *matches[i][1]*.

matches = matchChains(struct1,struct2,seqid=75,overlap=80)

Let's print out the information stored in matches. To do this, we first declare a function *printMatch*, which will print the sequence identity/overlap and RMSD of two polypeptide chains. Then, we have the code that calls this function for each of the rows in *matches*.

for match in matches: printMatch(match)

You can run the notebook by clicking on a block of code and clicking "run". Do this for every block of code starting at the beginning up to the preceding code. You will see that *matches[0]* is Chain A from *swiss_model_1* as well as Chain A from 6vxx, *matches[1]* is Chain A from *swiss_model_1* against Chain B from 6vxx, *matches[2]* is Chain A from *swiss_model_1* against Chain C from 6vxx, and then *matches[3]* starts over with Chain B from *swiss_model_1* against Chain A from 6vxx.

Exercise: When you ran *matchChains*, you most likely have in the output that the function failed to match chains initially and only succeeded when it used local sequence alignment. Why do you think this was the case? Explain your reasoning. (Hint: notice the lengths of the chains.)

> The sequences greatly differ in the number and length of chains (many indels present). Since global alignment tries to align the whole sequence at once, many insertions in the model sequence would be necessary to align the right areas, resulting in a very bad score. Local alignment allows regions to be aligned wherever

they are most compatible, so if there is say are matching domains between part A of the model and part J of the known structure, but also between part B of the model and part Z of the structure, these regions are allowed to separate and find their best-scoring match.

We want to match Chain A against Chain A (*matches[0]*), Chain B against Chain B (*matches[4]*), and Chain C against Chain C (*matches[8]*). That having been said, SARS-CoV-2 is a homotrimer, with all three chains very similar in structure, and the analysis below will hold up if we use another element of matches.

First, we will calculate the RMSD of Chain A from both proteins. This is satisfied by the following lines.

first_ca = matches[0][0] second_ca = matches[0][1] calcRMSD(first_ca, second_ca)

Run these lines. You will see that the RMSD of the two chains is very high. This is because we have not yet applied the Kabsch algorithm to ensure that the structures are correctly aligned. This alignment is provided by the *calcTransformation.apply()* function, which we now apply before recomputing RMSD.

calcTransformation(first_ca, second_ca).apply(first_ca); calcRMSD(first_ca,second_ca)

Now, run this code, which should give you a more reasonable value of RMSD.

Finally, we will combine all three chains of both our predicted structure and the verified structure, and then we will apply the Kabsch algorithm and as well as calculate the RMSD of the resulting protein.

first_ca = matches[0][0] + matches[4][0] + matches[8][0] second_ca = matches [0][1] + matches[4][1] + matches[8][1] calcTransformation(first_ca, second_ca).apply(first_ca); calcRMSD(first_ca, second_ca)

Run this code, and you have completed the notebook for this model.

Exercise: Provide a .zip file containing three screenshots. The first screenshot should show your finishing this part of the code to obtain an RMSD value between *swiss_model_1.pdb* and *6vxx.pdb*. Then, re-run the entire code block-by-block for *swiss_model_2.pdb*, and then re-run the code for *swiss_model_3.pdb*. Provide two more screenshots showing that you have found the RMSD values between the entire proteins of these two models and the experimentally validated SARS-CoV-2 spike protein.

Exercise: Change your Jupyter notebook so that you compare the experimentally validated structure of SARS-CoV-2 (6vxx) against the experimentally validated structure of SARS-CoV (6crx). What RMSD do you find? Would you say that SWISS-MODEL did a good enough job at predicting the structure of the SARS-CoV-2 protein? Explain your answer.

> calcTransformation kills the kernel for every combination of pdb files I tried. None of the RMSD values could be obtained this way. Does the assignment not work anymore, or am I missing something?
>
> If RMSD for the model is close to RMSD for the experimentally validated structure, then we woould say that SWISS-MODEL did a good job predicting the structure. If the RMSD is much higher though, the model did a poor job.

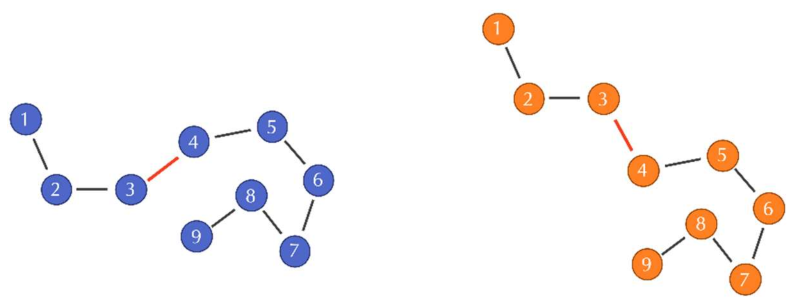
Comparing the SARS-CoV and SARS-CoV-2 spike protein structures locally

At the end of the last section, we compared the validated SARS-CoV and SARS-CoV-2 spike proteins. However, having just a single RMSD value does not greatly help us understand what the individual changes have been that have differentiated the SARS-CoV-2 spike protein. That is, we need to move toward comparing two structures locally.

Recall the following definition of RMSD for two protein structures *s* and *t*, in which each structure is represented by the positions of its *n* alpha carbons ($s1$, ..., sn) and ($t1$, ..., tn).

$$\mathrm{RMSD}(\mathbf{s},\mathbf{t}) = \sqrt{\frac{1}{n} \cdot \left(d(s_1,t_1)^2 + d(s_2,t_2)^2 + \cdots + d(s_n,t_n)^2\right)}$$

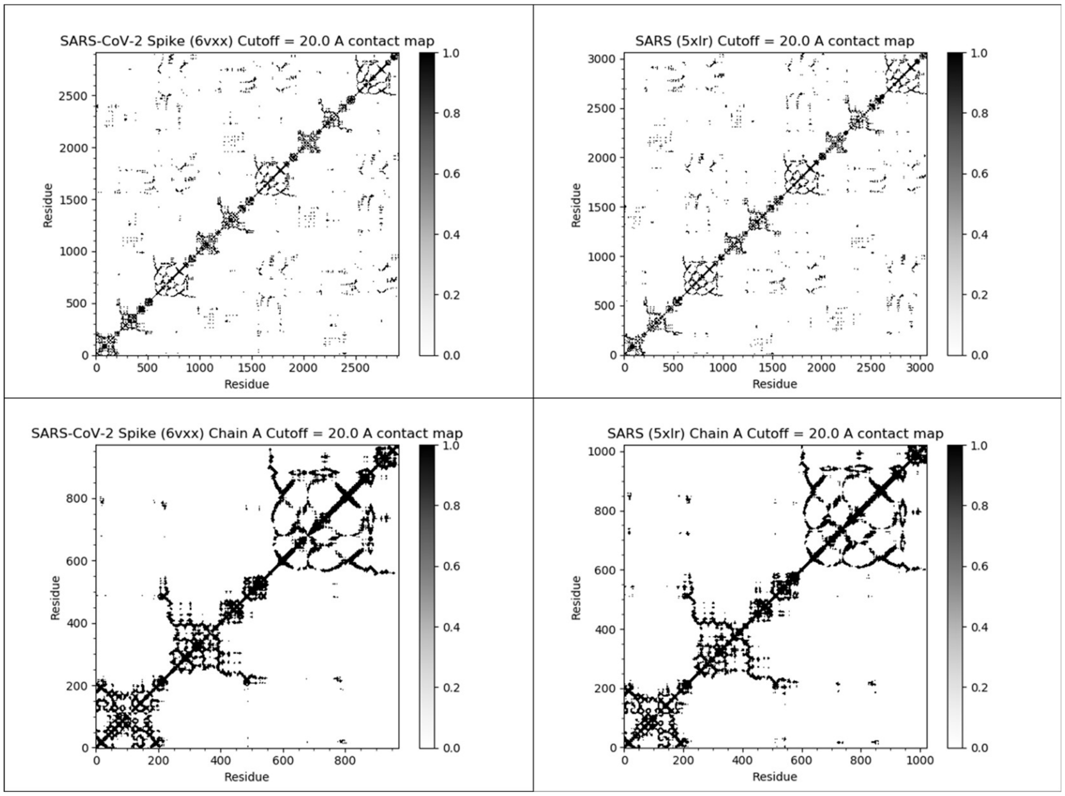
If two similar protein structures differ in a few locations, then the corresponding alpha carbon distances $d(si, ti)$ will likely be higher at these locations. However, one of the weaknesses of RMSD is that a change to a single bond angle at the *i*-th position may cause $d(sj, tj)$ to be nonzero when $j > i$, even though the structure of the protein downstream of this bond angle has not changed. For example, consider the following figure of two protein structures that are identical except for a single bond angle. All of the alpha carbon distances $d(si, ti)$ for *i* at least equal to 4 will be thrown off by this changed angle.



However, note that when *i* and *j* are both at least equal to 4, the distance $d(si, sj)$ between the *i*-th and *j*-th alpha carbons in *S* will still be similar to the distance $d(ti, tj)$ between the same alpha carbons in *T*. This observation leads us to a more robust approach for measuring differences in two protein structures, which compares *all* pairwise intraprotein distances $d(si, sj)$ in one protein structure against the corresponding distances $d(ti, tj)$ in the other structure.

To help us visualize all these pairwise distances, we will introduce the **contact map** of a protein structure *s*, which is a binary matrix indicating whether two alpha carbons are near each other in *s*. After setting a threshold distance, we set $M(i, j) = 1$ if the distance $d(si, sj)$ is less than the threshold, and we set $M(i, j) = 0$ if $d(si, sj)$ is greater than or equal to the threshold.

The figure below shows the contact maps for the SARS-CoV-2 and SARS-CoV spike proteins (both full proteins and single chains) with a threshold distance of twenty angstroms. In this map, we color contact map values black if they are equal to 1 (close amino acid pairs) and white if they are equal to 0 (distant amino acid pairs).



We observe two facts about these contact maps. First, many black values cluster around the main diagonal of the matrix, since amino acids that are nearby in the protein sequence will remain near each other in the 3-D structure. Second, the contact maps for the two proteins are very similar, reinforcing that the two proteins have similar structures.

We obtain some insight into how two proteins differ structurally at the *i*-th amino acid if we examine the values in the *i*-th row of the proteins' contact maps; that is, if we compare all of the $d(si, sj)$ values to all of the $d(ti, tj)$ values. In practice, researchers combine all of this information into a single metric called **Q per residue (Qres)** measuring the similarity of two structures at the *i*-th amino acid. The formal definition of Qres for two structures *s* and *t* having *N* amino acids is

$$Q_{res}^{(i)} = \frac{1}{N-k} \sum_{j \neq i-1, i, i+1}^{N} \exp[-\frac{[d(s_i, s_j) - d(t_i, t_j)]^2}{2\sigma_{i,j}^2}] .$$

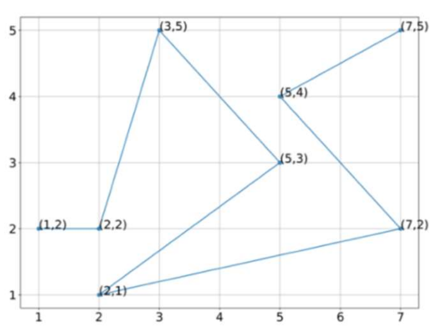
In this equation, *exp(x)* denotes *e* to the power of *x*. This equation also includes the following parameters.

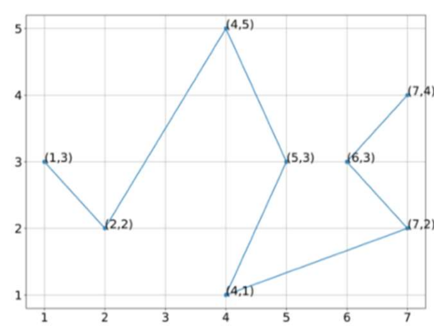
- k is equal to 2 at either the start or the end of the protein (i.e., i is equal to 1 or N), and k is equal to 3 otherwise.
- The variance term $\sigma2ij$ is equal to $|i-j|$^0.15, which corresponds to the sequence separation between the i-th and j-th alpha carbons.

If two proteins are very similar at the i-th alpha carbon, then for every j, the difference $d(si, sj) - d(ti, tj)$ will be close to zero, meaning that each term inside the summation in the Qres equation will be close to 1. The sum will be equal to approximately $N - k$, and so Qres will be close to 1. As two proteins become more different at the i-th alpha carbon, then the term inside the summation will head toward zero, and so will the value of Qres.

Qres is therefore a metric of similarity ranging between 0 and 1. Low scores indicate low similarity between two proteins at the i-th position, and higher scores indicate high similarity at this position.

Exercise: Consider the following figures reproduced from above. If the first sampled point in the two figures are, respectively, (1,2) and (1,3), compute the Qres of the fourth sampled point. Show your work to receive full credit.





j	s_x	s_y	t_x	t_y	d_s4_sj	d_t4_tj	var	Q(ij) res	include
1	1	2	1	3	3.606	3.606	1.179	1.000	Y
2	2	2	2	2	3.162	3.606	1.110	0.915	Y
3	3	5	4	5	0.000	0.000	1.000		N
4	5	3	4	1	2.828	4.000	0.000		N
5	5	4	5	3	2.236	2.236	1.000		N
6	7	2	6	3	5.000	2.828	1.110	0.119	Y
7	7	5	7	4	4.000	3.162	1.179	0.743	Y
							Q(i) res =	0.694	

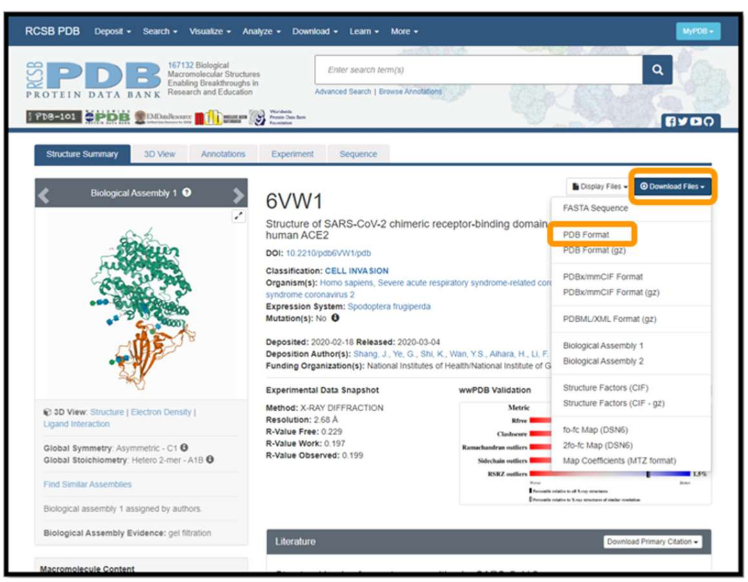
We now will compute Qres for the SARS-CoV and SARS-CoV-2 spike proteins, limiting our attention to a highly variable subregion of these proteins called the **receptor binding domain (RBD)**. We will use the VMD plugin *Multiseq*, a bioinformatics analysis environment, to align the SARS-CoV-2 RBD and SARS RBD having the PDB entries 6vw1 and 2ajf, respectively. After calculating Qres at every position, we will identify where the two RBD regions differ and visualize these regions within the larger protein structures.

It is important to note that the experimentally verified SARS-CoV-2 structure is a *chimeric* protein formed of the SARS-CoV RBD in which the RBM has the sequence from SARS-CoV-2. A chimeric RBD was used for complex technical reasons to ensure that the crystallization process during X-ray crystallography could be borrowed from that used for SARS-CoV.

Multiseq aligns two protein structures using a tool called **Structural Alignment of Multiple Proteins (STAMP)**. Much like the Kabsch algorithm discussed in class, STAMP minimizes the distance between alpha carbons of the aligned residues for each protein or molecule by applying rotations and translations. If the structures do not have common structures, then STAMP will fail. If you are interested in more details on the algorithm used by STAMP, click here.

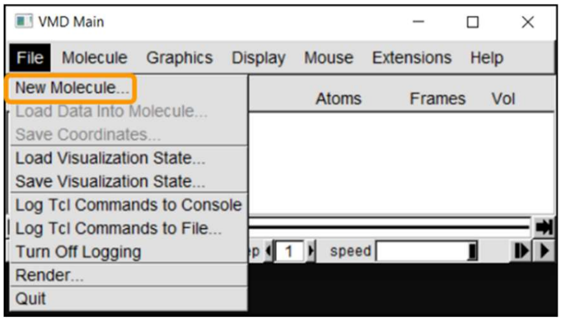
First, download VMD. In what follows, the program may prompt you to download additional protein database information, which you should accept.

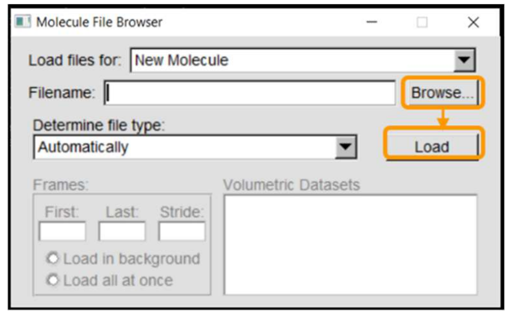
We will need to download the .pdb files for 6vw1 and 2ajf. Visit the 6vw1 and 2ajf PDB pages. For each protein, click on *Download Files* and select *PDB Format*. The following figure shows this for 6vw1.



Next, launch VMD, which will open three windows. We will not use *VMD.exe*, the console window, in this tutorial. We will load molecules and change visualizations in *VMD Main*, and we will use *OpenGL Display* to display our visualizations.

We will first load the SARS-CoV-2 RBD (6vw1) into VMD. In *VMD Main*, go to *File* > *New Molecule*. Click on *Browse*, select your downloaded file (*6vw1.pdb*) and click *Load*.





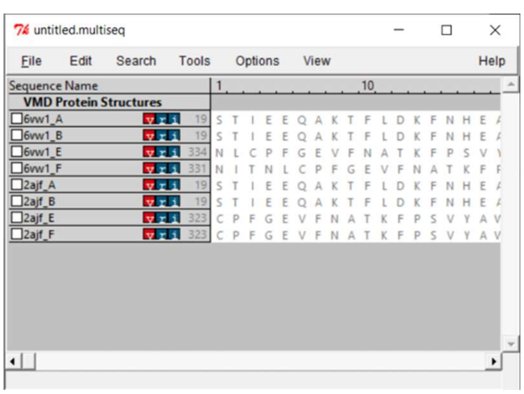
The molecule should now be listed in *VMD Main*, with its visualization in *OpenGL Display*.

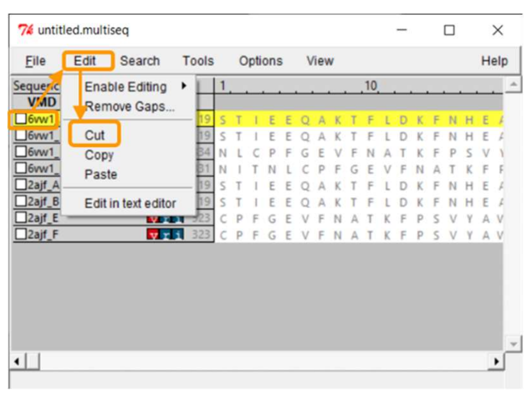
In the *OpenGL Display* window, you can click and drag the molecule to change its orientation. Pressing 'r' on your keyboard allows you to rotate the molecule, pressing 't' allows you to translate the molecule, and pressing 's' allows you to enlarge or shrink the molecule (or you can use your mouse's scroll wheel). Note that left click and right click have different actions.

We now will need to load the SARS-CoV RBD (2ajf). Repeat the above steps for *2ajf.pdb*.

After both molecules are loaded into VMD, start Multiseq by clicking on *Extensions* > *Analysis* > *Multiseq*.

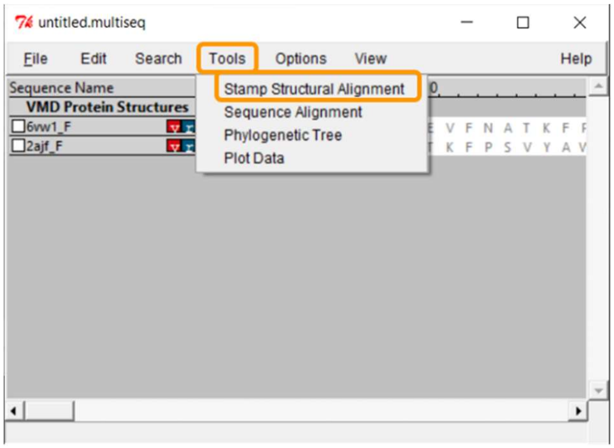
You will see all the chains listed out per file. Both .pdb files contain two biological assemblies of the structure. The first is made up of Chain A (ACE2) and Chain E (RBD), and the second is made up of Chain B (ACE2) and Chain F (RBD). Because Chain A is identical to Chain B, and Chain E is identical to Chain F, we only need to work with one assembly. (We will use the second assembly.)

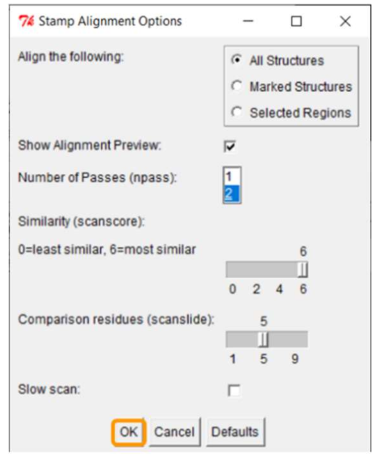




Because we only want to compare the RBD, we will only keep chain F of each structure. To remove the other chains, select the chain and click *Edit* > *Cut*. You can remove the nucleic structures as well.

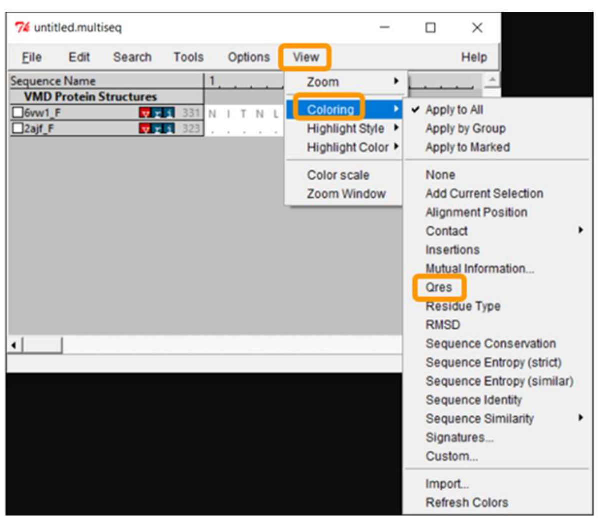
The protein structures are visible, but we need to overlap them in order to compare them visually. Click *Tools* > *Stamp Structural Alignment*, and a new window will open up.





Keep all the default settings and click OK; once you have done so, the RBD regions will have been aligned into a single structure.

Now that we have aligned the two RBD regions, we would like to compare their Qres values over the entire RBD. To see a coloring of the protein alignment based on Qres, click *View* > *Coloring* > *Qres*.



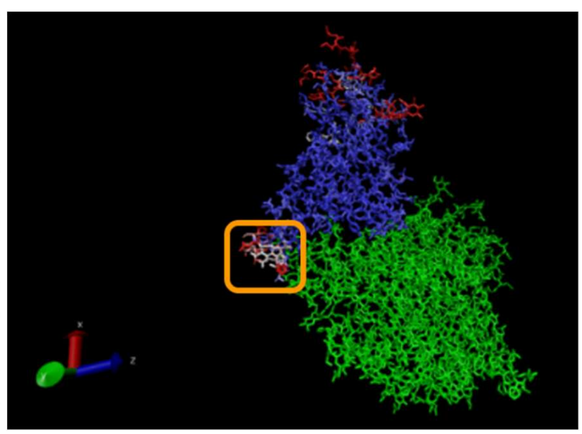
Scroll across the sequence alignment (in the *untitled.multiseq* window) to see how Qres changes over positions. Blue indicates a high value of Qres, meaning that the protein structures are similar at this position; red indicates low Qres and dissimilar protein structures.

If we zoom in on the region around position 150 of the alignment, then we can see a 13-column region of the alignment within the RBD region for which Qres values are significantly lower than they are elsewhere. This region, which is shown in the figure below, corresponds to positions 476 to 486 in the SARS-CoV-2 spike protein and positions 463 to 472 in the SARS-CoV genome.

The *OpenGL Display* window will now color the superimposed structures according to the value of Qres. We are interested in regions of the alignment having low Qres, which correspond to locations in which the coronavirus RBDs differ structurally. Note that these regions occur on the boundaries of the RBD's structure, where bonding is more likely to occur.

Exercise: There is another region in the interior of the RBD structural alignment that shows significant differences between the two protein structures. Where is it?

> Positions 383-394 of the SARS-Cov-2 spike protein and 370-282 of SARS-CoV do not align.



We won't ask you to do this, but we can color the alignment of the proteins complexed with ACE2 according to Qres, which produces the figure below. (The ACE2 enzyme is shown in green.) The low-Qres region of the RBM alignment that we highlighted in the above figure is outlined in the figure above.

Quantifying the strength of protein binding

We found regions of local structural differences between the SARS-CoV and SARS-CoV-2 spike proteins, but so what? We know that sequence differences at the level of DNA can wind up not influencing the amino acid sequence, and in the same spirit, we should not immediately conclude that a structural change in a protein will affect its function. How, then, can we *quantify* the affect of this structural change?

Earlier, we searched for the tertiary structure that best "explains" a protein's primary structure by looking for the structure with the lowest potential energy (i.e., the one that is the most chemically stable). In part, this means that we were looking for the structure that incorporates many attractive bonds present and few repulsive bonds.

To quantify whether two molecules bond well, we will borrow this idea and compute the potential energy of the complex formed by the two molecules. If two molecules bond together tightly, then the complex will have a very low potential energy. In turn, if we compare the SARS-CoV-2 RBD-ACE2 complex against the SARS-CoV RBD-ACE2 complex, and we find that the potential energy of the former is significantly smaller, then we can conclude that it is more stable and therefore bonded more tightly. This result would provide strong evidence for increased infectiousness of SARS-CoV-2.

We will compute the energy of the bound spike protein-ACE2 complex for the two viruses and see how the three regions we identified in the previous lesson contribute to the total energy of the complex. To do so, we will employ NAMD, a program that was designed for high-performance large system simulations of biological molecules and is most commonly used with VMD via a plugin called NAMD Energy. This plugin will allow us to isolate our specific region of interest to evaluate how much this local region contributes to the overall energy of the complex.

We will show how to use NAMD Energy to calculate the interaction energy for a bound complex, as well as to determine how much a given region of this complex contributes to the overall potential energy. We will use the chimeric SARS-CoV-2 RBD-ACE2 complex (PDB entry: 6vw1) and compute the interaction energy contributed by the site that we identified as a region of structural difference earlier.

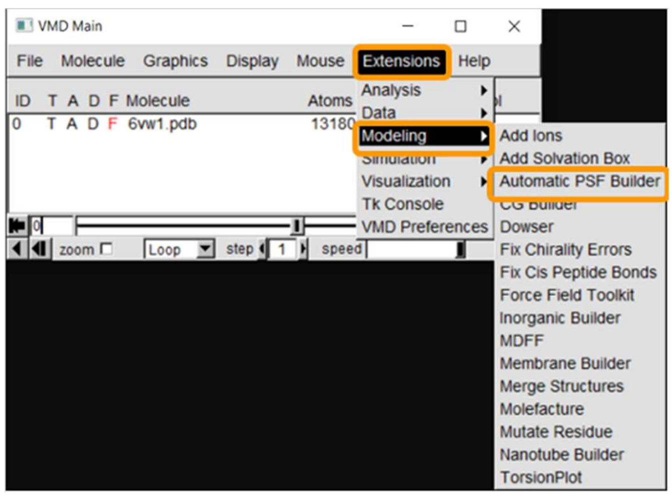
To determine the energy contributed by a region of a complex, we will need a "force field", an energy function with a collection of parameters that determine the energy of a given structure based on the positional relationships between atoms. There are many different force fields depending on the specific type of system being studied (e.g. DNA, RNA, lipids, proteins). There are many different approaches for generating a force field; for example, [Chemistry at Harvard Macromolecular Mechanics (CHARMM)]() offers a popular collection of force fields.

To get started, download [NAMD](), and remember where you place it in a directory on your computer.

NAMD needs to utilize the information in the force field to calculate the potential energy of a protein. To do this, it needs a **protein structure file (PSF)**. A PSF, which is molecule-specific, communicates between a given force field and a given molecular system. (See https://www.ks.uiuc.edu/Training/Tutorials/namd/namd-tutorial-unix-html/node23.html for more details.) Fortunately, there are programs that can generate a PSF given a force field and a .pdb file containing a protein structure. See [this NAMD tutorial]()

if you are interested in more information.

First, load [6vw1]() into VMD. We then need to create a protein structure file of 6vw1 to simulate the molecule. We will be using the VMD plugin *Automatic PSF Builder* to create the file. From VMD Main, click *Extensions* > *Modeling* > *Automatic PSF Builder*.

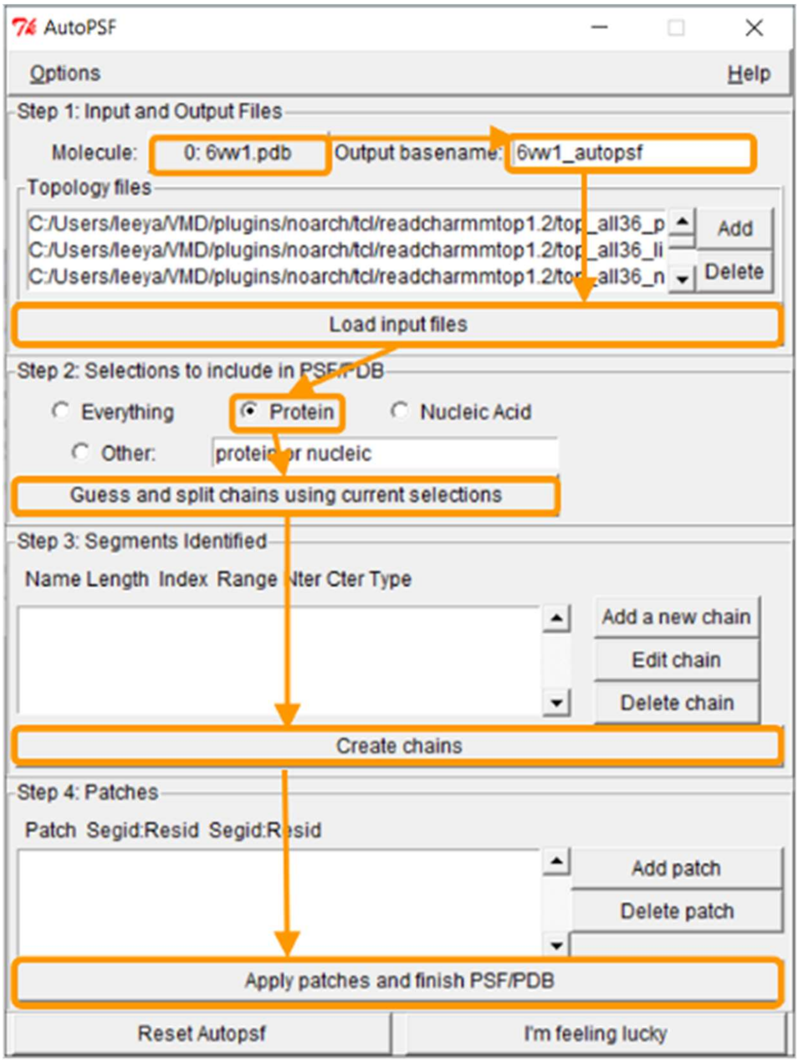


In the *AutoPSF* window, make sure that the selected molecule is *6vw1.pdb* and that the output is *6vw1_autopsf*. Click *Load input* files. In step 2, click Protein and then *Guess and split chains using current selections*. Then click *Create chains* and then *Apply patches and finish PSF/PDB*.

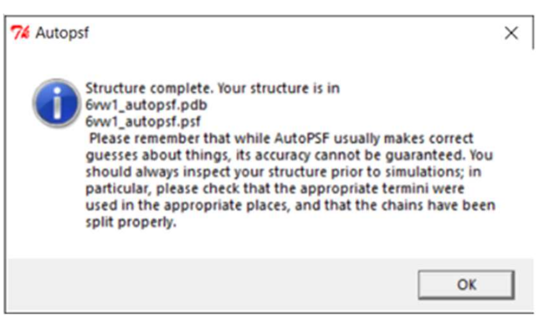
Notes:

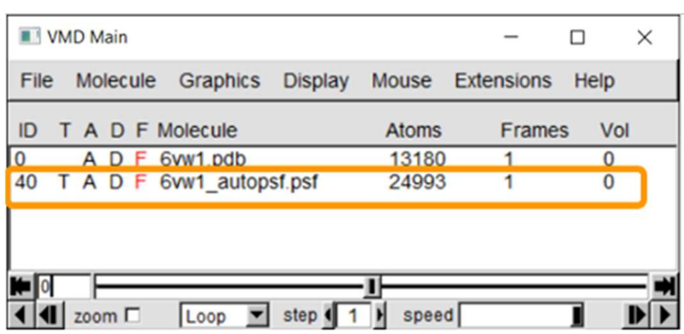
1. It may be the case that NAMD hangs when attempting to guess and split chains. If so, we are providing the PSF files here.
2. During this process, you may see an error message stating MOLECULE DESTROYED. If you see this message, click Reset Autopsf and repeat the above steps. The

selected molecule will change, so make sure that the selected molecule is 6vw1.pdb when you start over. Failed molecules remain in VMD, so deleting the failed molecule from VMD Main is recommended before each new attempt.



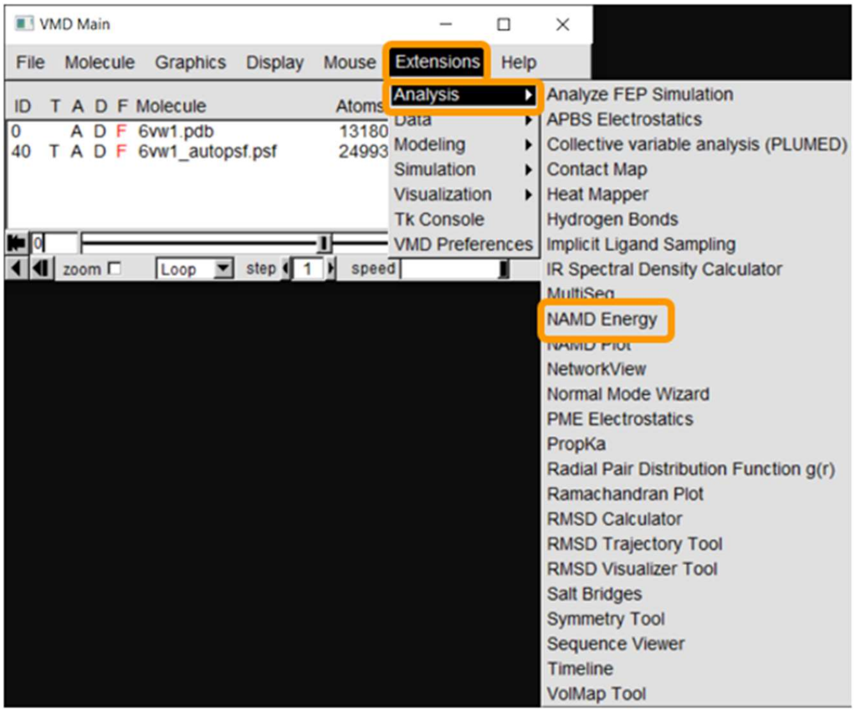
If the PSF file is successfully created, then you will see a message stating Structure complete. The VMD Main window also will have an additional line.





Note: If you were not able to generate the PSF file, load *both* the *6vw1_autopsf.psf* and *6vw1_autopsf.pdb* files into VMD at this point as new molecules.

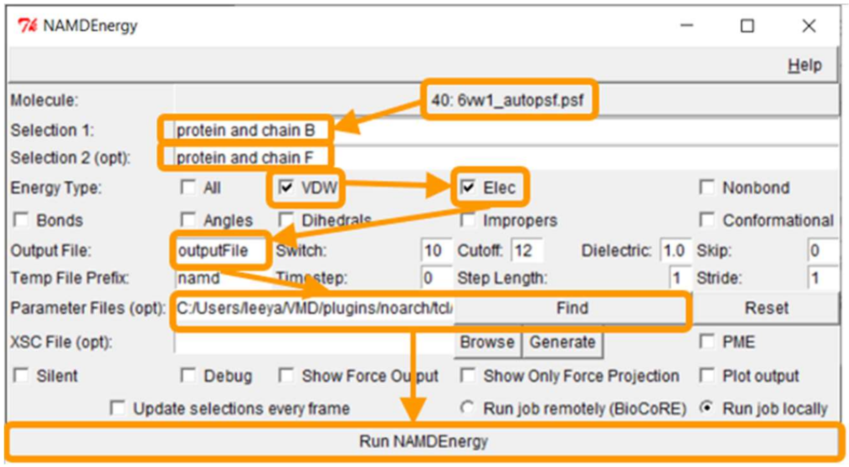
Now that we have the PSF file, we can proceed to NAMD Energy. In *VMD Main*, click *Extensions > Analysis > NAMD Energy*. The NAMDEnergy window will pop up. First, change the molecule to be the PSF file that we created.



The file 6vw1.pdb contains two biological assemblies of the complex. The first assembly contains Chain A (taken from ACE2) and chain E (taken from RBD), and the second assembly contains chain B (ACE2) and chain F (RBD). We will focus on the second assembly. Put protein and chain B and protein and chain F for Selection 1 and Selection 2, respectively.

Next, we want to calculate the main protein-protein interaction energies, divided over electrostatic and van der Waals forces. To do so, under *Energy Type*, check “VDW” and “Elec”.

Under *Output File*, enter your desired name for the results (e.g., *SARS-2_RBD-ACE2_B_F*). Next, we need to give NAMDEnergy the parameter file *par_all36_prot.prm*. This file should be found in your VMD installation at *VMD > plugins > noarch > tcl > readcharmmpar1.3 > par_all36_prot.prm*. (On a Mac you will need to right click on the VMD application in your Applications folder and click “Show package contents” to find these directories.) Finally, click *Run NAMDEnergy*. Note that you may be prompted by VMD for the location of your NAMD installation; if so, navigate to the directory where you placed NAMD and point VMD at the *NAMD.exe* file.



Note: If you were unable to generate the PSF file previously, then run the steps above first on *6vw1_autopsf.psf*, and if this fails, try running them on *6vw1_autopsf.pdb*.

The output file will be created in your current working directory (which may be your home directory if you have not set it in VMD) and can be opened with a simple text-editor. The values of your results may vary slightly upon repetitive calculations, but your results should be similar to the following. The energies here are negative because in physics, a negative value indicates an attractive force between two molecules, and a positive value indicates a repulsive force.

Frame	Time	Elec	VdW	Total
0	0	-149.81	-70.544	-220.354

We will now compute the interaction energy contributed only by between the site of interest that we found in SARS-CoV-2 RBD loop site (residues 476 to 486) and ACE2. In the NAMDEnergy window, enter *protein and chain B* for Selection 1 and *protein and chain F and (resid 476 to 486)* for Selection 2. Keep all other settings the same.

At the close of this assignment, we ask you to compare the bonded energy between the spike protein and the human ACE2 enzyme for SARS-CoV and SARS-CoV-2. Is it the case that SARS-CoV-2 is better bound? What about in the specific regions of interest that we identified? We leave the answer to you!

Exercise: Compute bonded energy in this manner for chains B and F for each of SARS-CoV and SARS-CoV-2 under three conditions: globally, our region of interest (residues 476 to 486 in SARS-CoV-2; residues 463 to 472 in SARS-CoV), and the range of residues that you found in the previous exercise. Provide the results of these six analyses.

Region	Electrostatic	van der Waals	Total
SARS-2: Global	-149.810	-70.544	-220.354
SARS-2: 476–486	-21.270	-8.690	-29.960
SARS-2: 383–394	0	0	0
SARS: Global	-130.661	-59.691	-190.352
SARS: 463–472	-20.737	-2.722	-23.459
SARS: 370–382	0	0	0

Exercise: Repeat the analysis from the previous exercise using chain A (ACE2) and chain E fsee (RBD).

Region	Electrostatic	van der Waals	Total
SARS-2: Global	-164.428	-73.822	-238.250
SARS-2: 476–486	-12.086	-9.690	-21.774
SARS-2: 383–394	0	0	0
SARS: Global	-95.273	-55.551	-150.964
SARS: 463–472	-13.234	-4.634	-17.868
SARS: 370–382	0	0	0

Exercise: Interpret the results of *global* protein analysis from the previous two exercises. How does the analysis differ depending on which chains are used? Is SARS-CoV-2 better bound to ACE2 or is SARS-CoV? (Hint: remember that negative energy means more tight bonding.)

> In both tables, SARS-2 has lower bonding energy than SARS. This means that SARS-2 binds to ACE2 better than SARS does.

Exercise: Interpret the results of *local* protein analysis from the previous two exercises regarding our two regions of interest. How do these two regions differ in terms of bonding energy?

> Region 1 (476-486 in SARS-2 and 463-472 in SARS) binds better than region 2 (383-394 of SARS-2 and 370-382 of SARS), which doesn't bind at all.

Conclusion: Bamboo shoots after the rain

Despite covering a great deal of ground in this challenge, we have left just as much unstudied. For one example, the surface of viruses and host cells are “fuzzy” because they are covered by structures called **glycans**, or numerous monosaccharides linked together. SARS-CoV and SARS-CoV-2 have a “glycan shield”, in which glycosylation of surface antigens allows the virus to hide from antibody detection. Researchers have found that the SARS-CoV-2 spike protein is heavily glycosylated, shielding around 40% of the protein from antibody recognition.

Although the study of proteins is in some sense the first major area of biological research to become influenced by computational approaches, this area is not just still active, but booming. With the COVID-19 pandemic reiterating the importance of biomedical research, the dawn of Alphafold showing the power of supercomputing to predict structure from sequence at unprecedented accuracy, and computational modeling promising to help find the medications of the future, new companies are rising up to study proteins like bamboo shoots after the rain.

Thus concludes our discussion of protein analysis. Thank you for joining us in these challenges!